Group Presentation

Use R for Statistics: What factors effect E-Commerce Sales

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Introduction

- E-commerce become an important part of business; necessity to find factors affecting the sales.
- Choose a set of sales data from Tianchi, an e-commerce platform database.
- Interested in factors: Sales Discount, COVID-19, and Customer Behavior.
- Randomly select a anonymized product to ensure privacy; regardless of specific product-related information.
- Use Measure of Location, t-Test, ANOVA, Linear Regression.

Data Preparation

R Output

```
spc_tbl_ [449 × 9] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ Date               : Date[1:449], format: "2019-05-30" ...
 $ count_view         : num [1:449] 196535 ...
 $ count_visitor      : num [1:449] 44672 ...
 $ avg_stay_time      : num [1:449] 112 ...
 $ bounce_rate        : num [1:449] 0.5 ...
 $ count_buyer        : num [1:449] 2994 ...
 $ count_deal         : num [1:449] 3159 ...
 $ money_deal         : num [1:449] 275033 ...
 $ count_deal_product: num [1:449] 10664 ...
```

`avg_stay_time`, and `count_deal_product` are not relevant to our project, so we just delete them.

Data Preparation *(Continued i.)*

Next we checked missing data.

`sum(is.na(data)) = 0`

And we should transform the column `date` from `'str'` to `'datetime'`.

```
data$Date <- as.Date(data$Date, format = "%Y/%m/%d")
```

What factors effect E-Commerce Sales

Descriptive Statistics

Stat Description

summary()

```
     Date              money_deal        count_deal      count_view      bounce_rate
 Min.   :2019-05-30   Min.   : 120489   Min.   : 1418   Min.   :142895   Min.   :0.3600
 1st Qu.:2019-09-19   1st Qu.: 388494   1st Qu.: 4746   1st Qu.:234470   1st Qu.:0.4400
 Median :2020-01-09   Median : 500704   Median : 6487   Median :275881   Median :0.4700
 Mean   :2020-01-09   Mean   : 538956   Mean   : 6959   Mean   :289573   Mean   :0.4816
 3rd Qu.:2020-04-30   3rd Qu.: 630138   3rd Qu.: 8519   3rd Qu.:327897   3rd Qu.:0.5100
 Max.   :2020-08-20   Max.   :1354130   Max.   :19944   Max.   :587284   Max.   :0.7400
```

We find some features

- mean is always larger than median, trend of right skewed;
- large range while small IQR, probably have outliers

Therefore, we should make measurements for the detection of shape and outliers.

Data: `Money_deal`

We greatly focus on `money_deal`.

Skewness

$$\mathrm{CS} = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^3}{\sigma^3} = 1.7 > 1$$

High degree of right skewed.

Kurtosis

$$\mathrm{CK} = \frac{\frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^4}{\sigma^4} = 4.53 > 3$$

Somewhat peaked; less dispersion; more outliers.

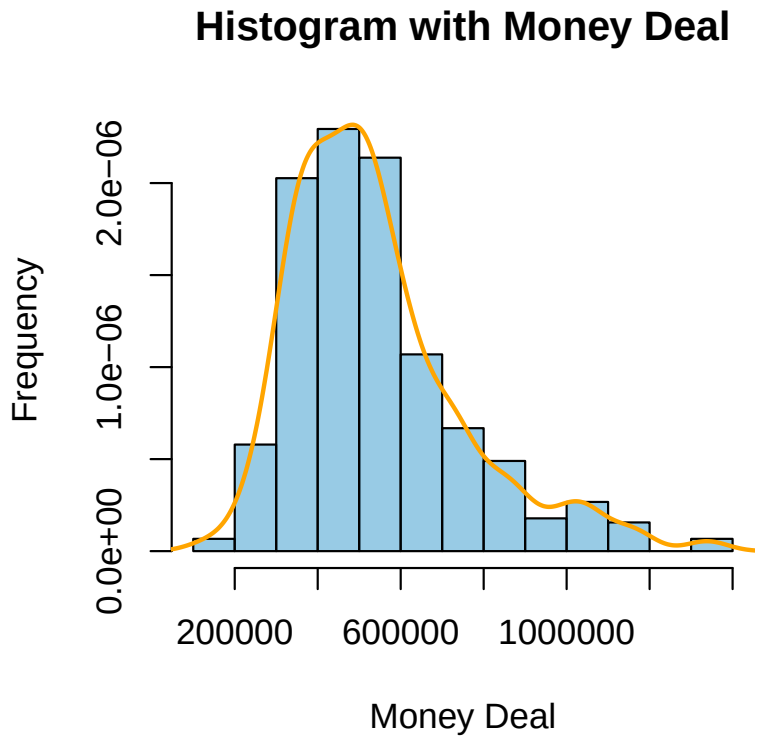


Figure: Histogram of `money_deal`

Outliers

Draw box-plot for outlier detecting.

Far-out Outliers

observations that less than lower fences or more than upper fence

$$x < Q_1 - k(Q_3 - Q_1)$$

or

$$x > Q_3 + k(Q_3 - Q_1)$$

where $k = 3$.

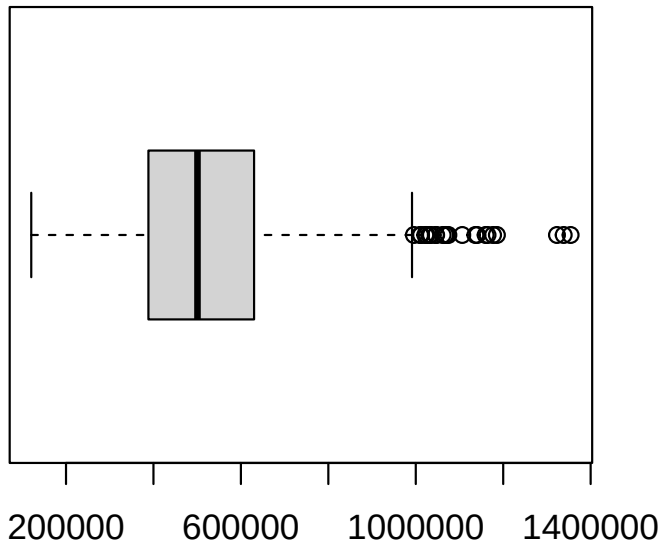


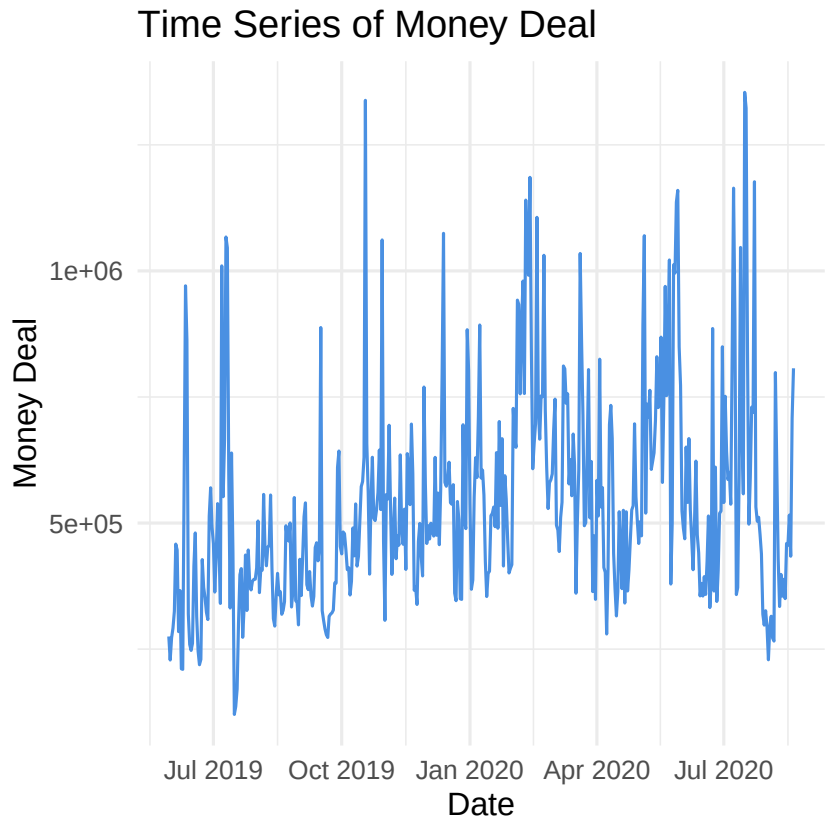
Figure: Boxplot of `money_deal`

What factors effect E-Commerce Sales

Factor about Sales Discount

Time Series

- 2 important period: 2019/11/11 and 2020/6/18
- guess discount rate effect the sale
- use paired t-test to examine



Student-t Test

Data Sampling

- x_1: 15 days sales before "618" event (include 6/18)
- x_2: 15 days sales before "1111" event (include 11/11)
- x_d: $x_1 - x_2$

Two-Sample Paired t-Test

Samples are dependent: two samples that have paired.

$$t = \frac{\bar{X}_D - \mu_0}{s_D/\sqrt{n}}$$

The degree of freedom used is $n - 1$.

Hypothesis

- Null Hypothesis H_0 : $\mu \geq 0$
- Alternative Hypothesis H_a $\mu < 0$
- Significant Level α: 0.05

t-Test Result

R Output

```
Paired t-test

data:  data_1$money_deal and data_2$money_deal
t = -2.199, df = 15, p-value = 0.02199
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
      -Inf -13550.61
sample estimates:
mean difference
      -66818.99
```

Evaluation

- $t = -2.199 < t_{\alpha;\, df=15} = -1.753,\ p-\text{value} = 0.02 < 0.05$
- sufficient evidence to conclude that sales of “618” is less than “1111”
- maybe due to the sales discount.

What factors effect E-Commerce Sales

Factor about the COVID-19

Analysis of Variance

Method

- divide `money_deal` into 5 seasons
- use ANOVA to examine the mean of each season

Hypothesis

- Null Hypothesis $H_0: \ \mu_1 = \mu_2 = \cdots = \mu_5$
- Alternative Hypothesis H_a: at least one mean is different
- Significant Level α: 0.05

ANOVA

$$F = \frac{MSA}{MSW} = \frac{SSA/\left(k-1\right)}{SSW/\left(n-k\right)} = \frac{\sum_{j=1}^{k} n_j \left(\bar{x}_j - \bar{\bar{x}}_j\right)^2 / \left(k-1\right)}{\sum_{j=1}^{k} \left(n_j - 1\right) s_j^2 / \left(n-k\right)}$$

ANOVA result

R Output

```
             Df    Sum Sq   Mean Sq F value   Pr(>F)
group         4 2.885e+12 7.212e+11   19.22 1.37e-14 ***
Residuals   442 1.659e+13 3.752e+10
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Evaluation

- $p-\text{value} < 0.05$
- sufficient evidence to prove H_a
- maybe due to COVID-19

ANOVA result *(Continued i.)*

Possible Reason

- hoarded goods in the early stage of COVID-19, `money_deal` increase rapidly;
- reduced consumption after the epidemic began, `money_deal` decrease dramatically.

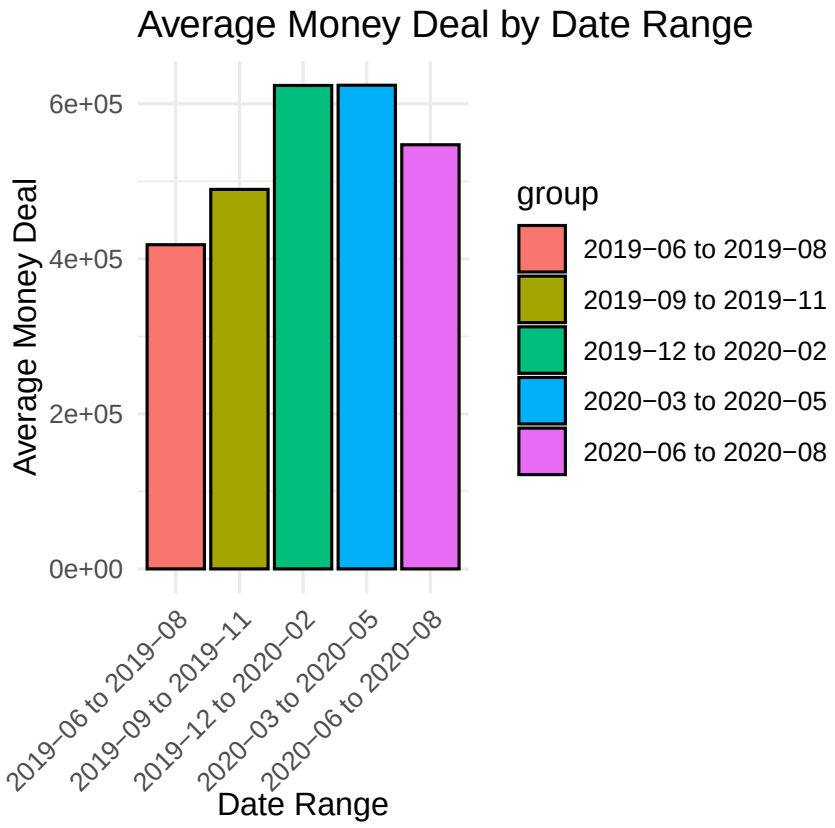


Figure: Season Mean

What factors effect E-Commerce Sales

Factor about Customer Behavior

Pairs & Correlation

Correlation Coefficient

$$\text{Cor}\left(r_{xy}\right) = \frac{\text{Cov}\left(X,\, Y\right)}{s_x s_y} = \frac{\sum_{i=1}^{n}\left(x_i - \bar{x}\right)\left(y_i - \bar{y}\right)}{\left(n-1\right) s_x s_y}$$

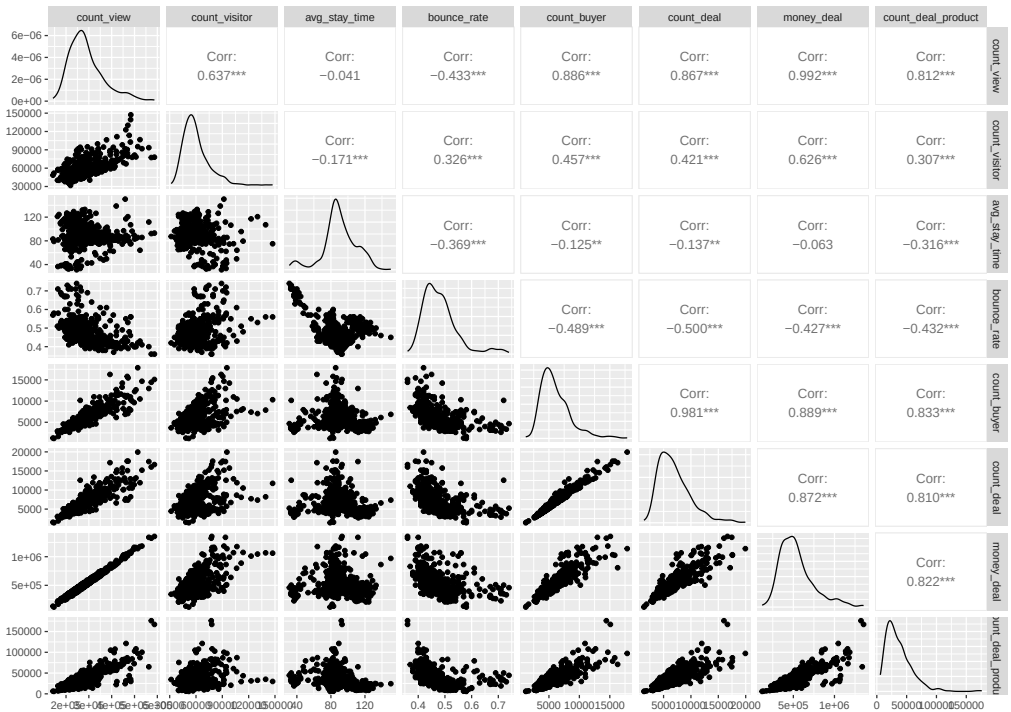


Figure: Paired Variables Plot/Correlation

Generalized Pairs *(Continued i.)*

Evaluation

- positive/negative effect: all parameters
- linear relationship: `count_view` and `money_deal`
- high correlation: `count_buyer`, `count_deal` and `count_view`

Conclusion

Use parameter `count_view` and `bounce_rate` and their product.

Linear Regression Model

```
Call:
lm(formula = money_deal ~ count_view * bounce_rate, data = df)

Residuals:
   Min     1Q Median     3Q    Max
-52308 -21709   2710  21290  52253

Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)             -3.270e+05  3.110e+04 -10.514  < 2e-16 ***
count_view               2.991e+00  1.072e-01  27.888  < 2e-16 ***
bounce_rate              1.739e+05  6.459e+04   2.693  0.00735 **
count_view:bounce_rate -6.111e-01  2.321e-01  -2.633  0.00875 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 25710 on 445 degrees of freedom
Multiple R-squared:  0.985,    Adjusted R-squared:  0.9849
F-statistic:  9758 on 3 and 445 DF,  p-value: < 2.2e-16
```

Check Assumptions

Assumptions

- Linearity: check the linear relationship between the dependent and independent variables.
- Normality: the residuals follow a normal distribution
- Homoscedasticity: variation about the regression line is constant
- Independence: observations are independent of each other

Assumption i: Linearity

Method

Check by plotting the residuals against the fitted values.

Evaluation

- plot appears random

Conclusion

There is a linear relationship.

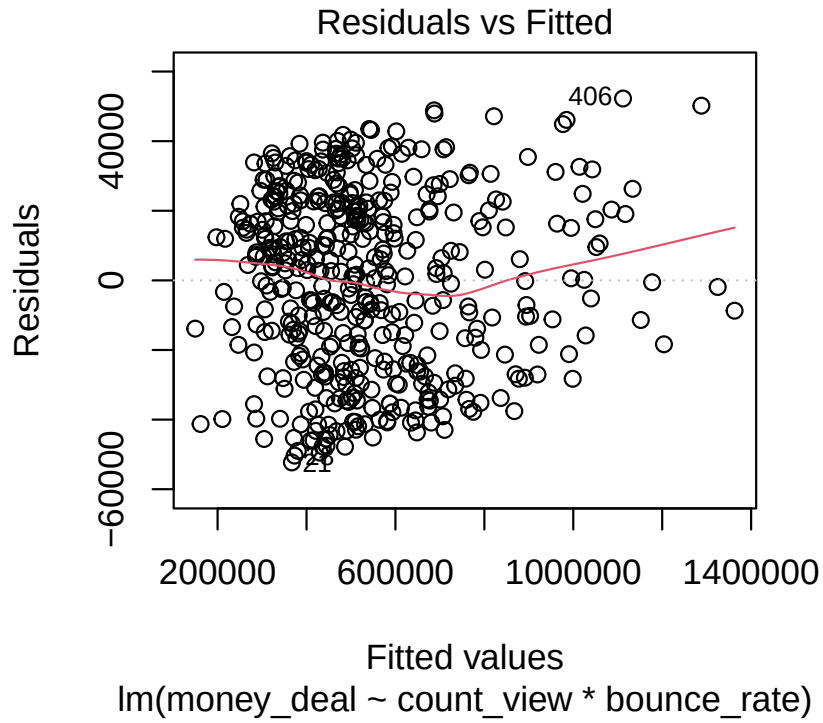


Figure: Residual of Model

Assumption ii: Homoscedasity

Method

Check by Non-constant Variance Test (ncvTest)

Output

```
ncvTest(model)

Non-constant Variance Score Test
Variance formula: ~ fitted.values
Chisquare = 0.4163145, Df = 1, p = 0.51878
```

Assumption iii: Normality

Method

Check by creating a Q-Q plot.

Evaluation

- points fall along a line

Conclusion

There is a normal distributed residual.

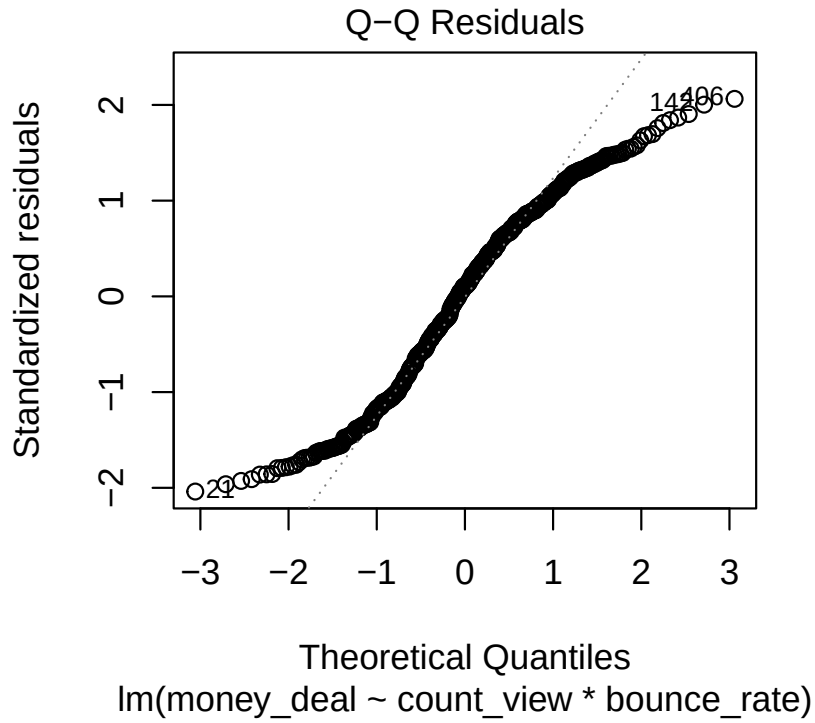


Figure: Q-Q Plot

Assumption iv: Independency

Method

Check by plotting the residual time series.

Evaluation

- no pattern or trend
- data is cross-sectional

Conclusion

The observations are independent of each other.

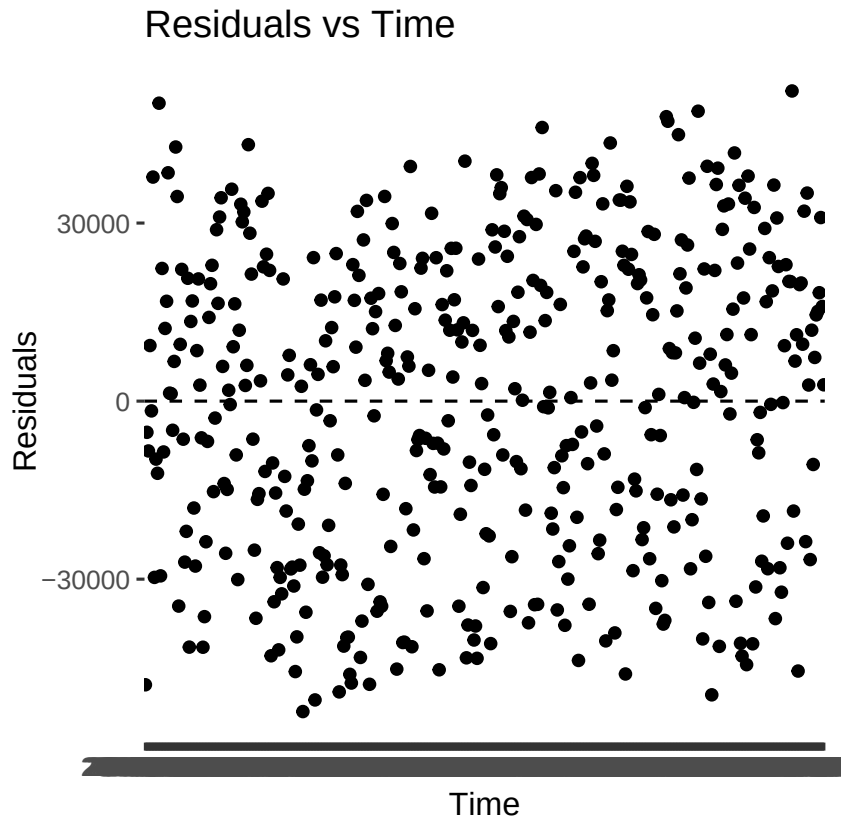


Figure: Time Series of Residual

Conclusion

Findings

- Sales depend on Season, COVID-19, Customer Behavior
- Sales could be predicted by the amount of view and bounce rate

Goal

- Find solution to stimulate Off-season
- Changing sales models to adapt to the new social environment